VISION BASED GOALKEEPER ROBOTIC ARM

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Introduction

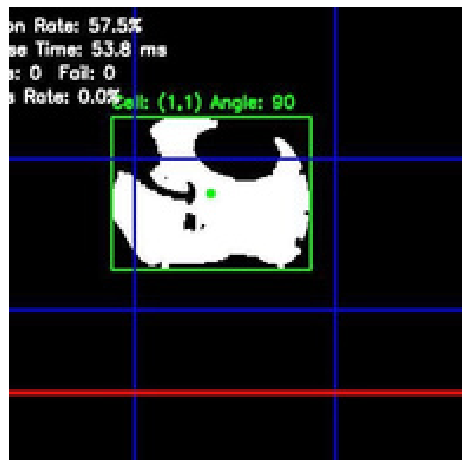
Real-time ball interception requires fast detection and accurate motor response. This project presents a vision-based goalkeeper robotic arm that detects a ball and moves a servo motor to block it using a grid-based control approach.eliable performance using a grid-based control approach.

Objective

To design a real-time vision-based system that detects a moving ball and controls a robotic arm to block it accurately.

Methodology

1. Capture image from camera
2. Detect ball using HSV color filtering
3. Convert to binary image
4. Divide frame into a 3×3 grid
5. Map detected position to servo angle
6. Send command to Arduino for actuation

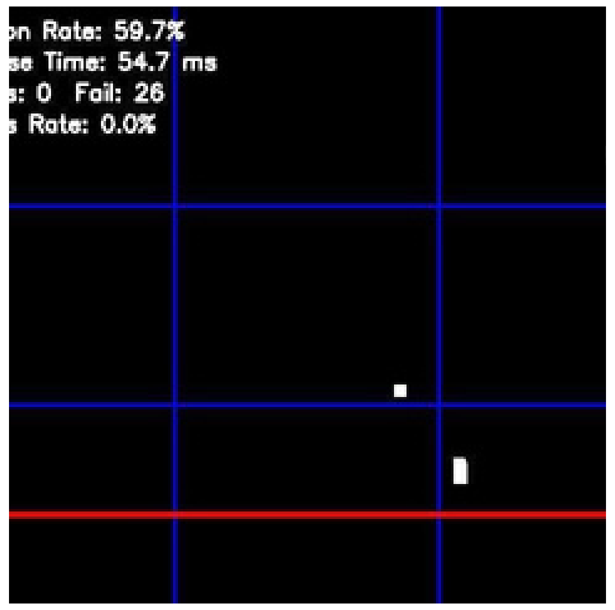


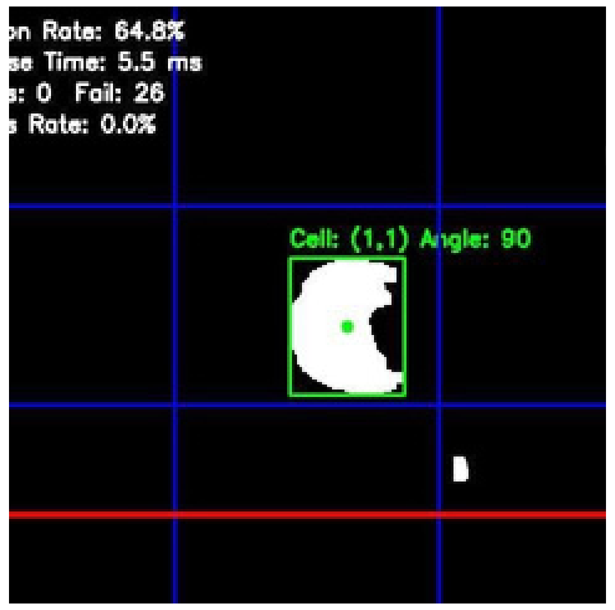
System Overview

The system uses a Raspberry Pi for image processing and an Arduino to control a servo motor based on the detected ball position.

Grid System

The frame is divided into a 3×3 grid, where each region corresponds to a predefined servo angle for fast decision making





KPIs

- Detection Rate: 59.7 %
- Response Time: 54.7 ms

Limitations

The system is sensitive to lighting conditions and may experience detection errors under fast motion or complex backgrounds.

Results

The system successfully detects the ball and tracks its position in real time. The binary processing highlights the detected object clearly, while the grid-based mapping allows fast and stable servo response.

Analysis

The analysis section interprets the results and discusses their implications. It explains the meaning of the findings in relation to the research objectives and existing literature. The analysis may involve comparing the findings to previous research, identifying patterns and trends, or drawing inferences. This section should provide a critical evaluation of the results and their significance.

Conclusion

The system successfully demonstrates real-time ball detection and robotic interception using vision-based control. The grid-based approach improves response stability and enables fast and reliable servo movement.

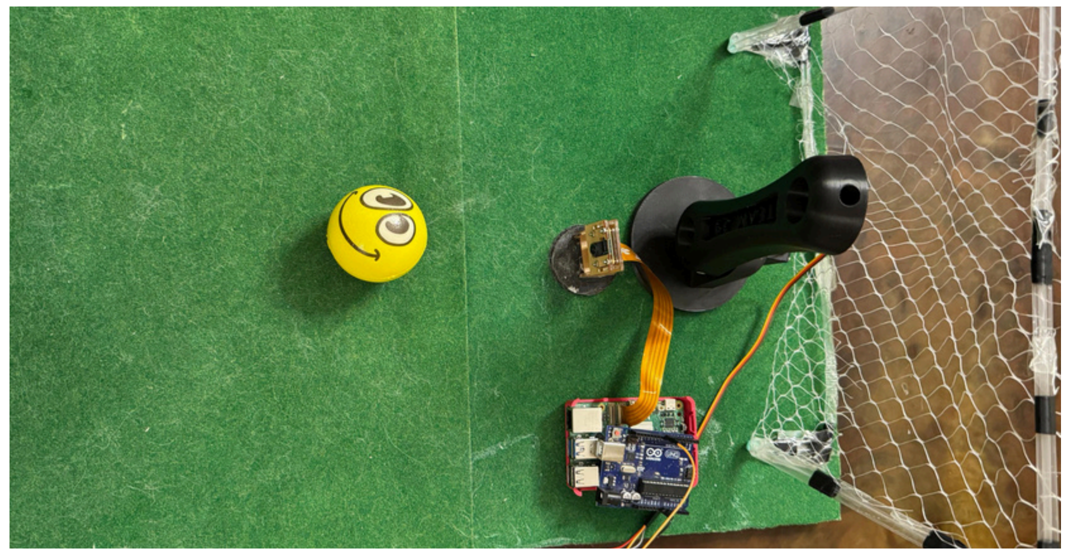


Image Processing for Mechatronics (MCTR-1010) – Spring ‘26 -- Dr. Eng. Omar M. Shehata